## Possible analyses

Seth's list from the reports, with the status of each item as of November 4. Pending means coded and waiting on the VRDC run. Reviewer 1's comments are at the end.

# 1 Data and descriptive

If worthwhile for some of these descriptive exercises, I have a PhD student who might be able to do them. We would have to get her access to the VRDC...

Status: No need to get another person involved. Plus it would cost $15k for her to have a seat! Most of this is easy enough to code up so I'm working on it..

Heterogeneity within DRG: distribution of patient risk scores.

Status: Pending

How well do the patient risk scores predict failure, net of the doc's observed performance?

Status: Pending

(R2 #3) Volume - outcome relationship. Does performance increase with volume (over long-term) within surgeons? Does it decrease (over short-term, eg within a year) when there is high congestion? Could estimate LPMs with patient risk-adjusters and surgeon FEs.

Status: Pending

CF reallocation and volume. Plot histogram of observed patient counts (e.g., per year), and plot in CF reallocation. Check the upper tails.

Status: Pending

Show that surgeon volume (and its heterogeneity across docs) is pretty consistent over time. Could compute within vs between variance in volume (e.g., by month or quarter) and compare the two. (From R2's 3b criticism of our long time periods for congestion.)

Status: Pending

Heterogeneity in patient char's across surgeons. Plot dist's across surgeons of key demographics: race, sex, age (e.g., use averages of these char's over the 6 year estimation period). Also dist of risk scores. Then est correlation of these features with the observed quality (i.e., performance over 11 years).

Status: Pending

Could we look at hospital discharge (or all payer) data in some state(s), to see how much volume we are capturing, and how much heterogeneity there is across surgeons? Concern is that best docs are at capacity (more than our descriptive analysis suggests) due to commercial patients. (R4 #5)

Status: Not started; a data request rather than code. HCUP state inpatient data with physician identifiers (Florida, New York, and a few others license the surgeon NPI) would give each surgeon's all-payer joint-replacement volume to set against the Medicare count.

## 2 More programming and computation

Other change to event study: DDD-ish comparing change in referrals (per fixed calendar time like a qtr) when see a failure vs when see a success. This is R2's idea to address mean reversion.

Status: Pending

Could a surgeon-level event study include spillover from surgeon A to surgeon B, if A is in B's group? (R2 #5)

Status: Pending

Should we also do a practice-level event study?

Status: Pending

Logit interacting patient risk scores with surgeon average quality (maybe leave-one-out or 5 year avg before patient t, not pair-specific but all patients). R2 suggested to interact with distance, but this interaction with observed performance seems better.

Status: Pending

Logit using PCP movers?

Status: Pending

Logit using relatively new PCPs?

Status: Pending

(above two could pool HRRs to increase power)

Status: Both pooled. Pending

Simulate the model for one market, using our estimated parameters. Program the full solution as well as Gittins. See how much the simulated behavior differs btw the two methods.

Status: Built locally in R (exact solution of the learning problem against the Gittins index); not yet run at scale.

Not now - maybe, depending on some descriptive assessments:

(Estimate structural model using practice groups. R4 #2and6)

Status: Pending, as a full parallel set at the practice level. The earlier run's sign flip on the existing-familiarity counterfactual is the thing to explain; the likely reason is within-group routing, and the new descriptives measure it.

(Include average effect of volume / surgeon learning in the structural model.)

Status: Not started. The within-surgeon LPM above is the descriptive check that would come first.

# 3 Theory

# 4 Other items from the reports

Items we have taken up that are not on the list above.

R1 #4–5, R2 #2, R3 2a (the event study).

Status: Done. The submitted event study selects treated pairs on a referral at the event quarter while comparison pairs are not; the replacement compares the PCP's next referral after a failure with the next referral after a success, in levels and as an event study in referral order with clean windows, with pair fixed effects and a split by the calendar gap to the next referral. The write-up is `updated-reduced-form.tex`.

R3 1a (the Gittins index with logit shocks).

Status: The simulation above is the expectations check; it isolates exactly the gap R3 describes, since the exact solution and the index solve the same learning problem with the same discount factor.

R2 3c, R3 1b (the counterfactual fixed point).

Status: Writing. The main text needs the equilibrium stated: what adjusts (each surgeon's volume, through the congestion term), the fixed point solved, and the uniqueness check already in the appendix.

R2 3b (congestion at three-year windows).

Status: Pending: yearly congestion in the markets large enough for specialist-by-year effects, within surgeon.

R2 minor 3 (convergence).

Status: Pending: every market's convergence is now logged with its size, with one retry and a likely cause removed.

R2 minor 4 (myopic against forward-looking).

Status: Pending: a Vuong test by market.

R2 minor 2 (established PCPs against all PCPs).

Status: Pending

# 5 Reviewer 1

1. The identification claim against Ching, Erdem, and Keane (2013).

Status: Writing. Drop the formal-versus-verbal contrast and minimize arguments about contribution related to identification.

2. DRG 470 and case mix.

Status: Already in the appendix (D.4, D.5)...not sure there's anything else to do here.

3. Risk aversion as an alternative reading of the familiarity effect.

Status: Can be mainly handled in the writing I think.

4. The gap between the event study (16%) and the structural partial effect, and the suggestion to estimate the model on the event-study sample.

Status: Already addressed in new reduced form analysis.

5. One failure should move a well-established posterior little; adaptive or recency-weighted learning.

Status: Partly addressed, partly writing. Not sure about "adaptive" or "recency-weighted"...

6. Whether surgeons' skills evolve over the panel.

Status: Incorporated as part of new analysis looking at within-surgeon probability of failure as a function of cumulative experience.

7. Positioning: intermediary learning is not new; patient-level outcomes together with the intermediary's choice is.

Status: Writing point. We could try to reframe the contribution around that combination if we want...